Education and Training in Autism and Developmental Disabilities, 2018, 53(4), 363–377
© Division on Autism and Developmental Disabilities

Effectiveness of Video Modeling Presented via Smartboard for Teaching Social Response Behavior to Children with Autism

Esin Pektaş Karabekir and Nurgül Akmanoğlu
Anadolu University

Abstract: This study aims to examine the effectiveness of video modeling presented via smartboard for teaching children with autism about social response in the form of gestures, and verbal and facial expressions indicative of happiness and anger, and evaluate the method in terms of maintenance and generalization. The study used a multiple probe design across participants. Three male subjects diagnosed with autism, ages 3 to 5, participated in the study. Results showed that with video modeling presented via smartboard, all three subjects were able to learn about social response in the form of gestures, verbal and facial expressions indicative of happiness and anger, and generalize their learning to different settings, materials and persons. In addition, maintenance sessions conducted in weeks four and six and month seven following the end of the instruction showed that subjects retained the learned behavior. Social validity results showed that mothers and teachers of the children had positive views about the study.

Children diagnosed with autism under DSM-5 have visible deficiencies in social communication and interaction (American Psychiatric Association-APA, 2013). Among the most visible deficits in this field include (a) deficits in initiating, engaging in or maintaining social interaction, (b) deficits or inability in making eye contact, using unusual gestures or facial expressions, and understanding and using gestures and facial expressions, and (c) deficits in dialogue and sharing interests and feelings (APA, 2013; Webber & Scheuermann, 2008). Limitations in the use of gestures and facial expressions, in particular, may prevent these individuals from engaging in successful social interaction, and create difficulties in forming and maintaining friendships with peers (Baron-Cohen, 1988; Leaf, Dotson, Oppenheim, Sheldon, & Sherman, 2010).

In child development, the ability to understand and express feelings in social contexts in which one is placed is an important concept. Giving the proper response both to discriminative stimuli in the social context and to affective behaviors of others is an important behavior for the development of interpersonal social interaction (Gena, Couloura, & Kymissis, 2005). This skill involves the ability to make connections between contexts, affective states and messages sent via facial and verbal expressions, intonation and gestures (Saymaz, 2008). These behaviors are also called social response behaviors. Social response behaviors include understanding and expressing emotions that have a social communicative function, and observable behaviors such as facial and verbal expressions, intonation and gestures (Buffington, Krantz, McClannahan, & Poulson, 1998; Gena, Krantz, McClannahan, & Poulson, 1996; Gena et al., 2005).

Facial expressions and gestures are among the most important social response behaviors. In children with autism, the development of gestures is affected by both qualitative and quantitative deficits. Children with autism have deficits in the natural and spontaneous use of gestures to express emotions depending on the social context, comprehending gestures used by others, and imitating gestures (Ingersoll, Lewis, & Kroman 2006; Plimley &

This study was supported by a Grant from Anadolu University Research Fund (Project No: 1508E574). Authors are grateful to NOVA Language Services for proof reading of the manuscript. Correspondence concerning this article should be addressed to Nurgül Akmanoğlu, Anadolu University, Research Institute for the Handicapped, Eskisehir, 26470, TURKEY. E-mail: nakmanoglu@anadolu.edu.tr

Bowen, 2007). Facial expressions, which play an important role in the development of social competence, are considered to be the primary and most important source of information mediated via social affective and verbal communication (Balconi, Amenta, & Ferrari, 2012; Tardif, Laine, Rodriguez, & Gepner, 2007). Facial expressions, when used properly, reflect emotions and allow the transmission of emotions appropriate for the social context via the face. Expressing emotions using specific facial expressions is considered to play a very important role in social interaction and communication (Gena et al., 1996). In addition, the social context that gives rise to the emotion is considered to have an important effect when assigning meaning to facial expressions and emotions. For a child to learn about the affective responses that are appropriate for the social context, the ability to perceive the reasons and conditions that give rise to the emotion in question, known as affective experience in the literature, is also important (Akmanoğlu, 2015; Balconi & Carrera, 2007; Balconi et al., 2012).

One of the well-known strengths of children with autism is that they are able to perceive visual stimuli more easily compared to other stimuli, and can use visual clues in a more meaningful way. In addition, some studies report that children with autism tend to be visual learners rather than auditory learners, and therefore watching videos is a pleasurable activity for them (Nikopoulos & Keenan, 2006; Quill, 1997; Schneider & Goldstein, 2010; Weiss, 2013). Because children with autism prefer visually presented stimuli and respond more positively to visual interventions, video modeling is an evidence based method used effectively to instruct children with autism (National Autism Center-NAC, 2015; The National Professional Development Center on Autism Spectrum Disorders-NPDC, 2014). Video modeling involves the child watching a video in which a model presents the target behavior or skill in an appropriate manner, and then independently performing this behavior by taking the behavior presented in the video as a model (Charlop-Christy, Le, & Freeman, 2000; Delano, 2007; Murray & Noland, 2013; NAC, 2015; Nikopoulos & Keenan, 2006). In a report published by the NAC (2015), modeling is considered to be effective in teaching higher order cognitive skills, academic skills, communication skills, interpersonal communication skills, interpersonal responsibility, play skills, problem behaviors, social affective skills and sensory control.

With the development of technology and easy access to devices, images used in video modeling started to be presented via various technological tools. The manner in which video modeling is presented depends, in addition to child characteristics and characteristics of the skill being taught, on the opportunities available to the trainer and the trainer's experiences and preferences. Today, video modeling is presented via portable and newer technologies such as tablet computers, smart phones, laptop computers, and smartboards (Argott, 2012).

A limited number of studies show that video modeling can be used effectively for teaching social response behaviors to individuals with autism (Akmanoğlu, 2015; Axe & Evans, 2012; Charlop-Christy & Daneshvar, 2003). We were able to find only a single study on the use of video modeling to teach all social response behaviors including, gestures, intonation, verbal and facial expressions. Charlop, Dennis, Carpenter, & Greenberg (2010) aimed to teach proper social response behaviors (gestures, intonation, verbal and facial expressions) during social interaction to three children with autism, using the video modeling method. In the study, video modeling was used to present the proper gestures, intonation, verbal and facial expressions during social interaction. The study was conducted with the participation of three children diagnosed with autism, ages 7 to 11. The study used a multiple baseline design across behaviors, a single subject design. Results of the study showed that video modeling was effective in teaching social response behaviors to all the participants with autism, and all subjects were able to generalize the acquired behaviors to different contexts, persons and stimuli. The study did not collect maintenance data.

These studies on the teaching of social response behaviors suggest that video modeling can be used to teach gestures, intonation, verbal and facial expressions, but they have some limitations. These limitations, which also justify the present study, are as follows: (a) In the one study on teaching all social response be-

haviors including gestures, verbal and facial expressions, social responses were taught without associating these responses with affective states or contexts related to these affective states (Charlop et al., 2010). This study (Charlop et al., 2010) did not include affective states associated with social responses or contexts in which emotions arise, which are known to be an important variable in the emergence of emotions and related social responses. Therefore, there is a need for studies that include social contexts associated with emotions that give rise to social responses. (b) As limitations of their study, Charlop et al. (2010) state that they did not collect generalization data on whether the social responses taught would be generalized to other contexts not included in the study, or maintenance data that would test the retention of the behaviors taught. Collecting generalization and maintenance data is important to remove doubts about the dependent variable (Kennedy, 2005). (c) A review of the literature shows that only about one third of the studies on the use of video modeling to teach social skills report social validity data. Failure to collect social validity data from the direct stakeholders in the study is an important limitation as it hinders the evaluation of the social acceptance of the behavior taught, the appropriateness of video modeling, and the significance of the effects (Acar & Diken, 2012; Banda, Copple, Koul, Sancibrian, & Bogschutz, 2010).

For these reasons, the present study aims to examine the effectiveness of video modeling presented via smartboard for teaching children with autism about social responses in the form of gestures, verbal and facial expressions indicative of happiness and anger. To this end, answers were sought to the following questions: (1) Is video modeling presented via a smartboard is effective in (a) the acquisition of social response behaviors in the form of verbal expression, use of gestures, and facial expressions indicative of the affective states of happiness and anger, (b) the maintenance of these behaviors 4 weeks, 6 weeks and 7 months after acquisition, and (c) the generalization of this skill to different people, different settings and different social contexts created using different materials? (2) What are the views of the parents and teachers of the children who participated in the study (social validity)?

Method

Participants

Subjects. The study was conducted with the participation of three male subjects diagnosed with autism by the child psychiatry clinic of a public hospital. All of the subjects attend group sessions from 09.00 a.m. to 12.30 p.m. in the Developmental Support Unit of Anadolu University's Research Institute for Individuals with Disabilities, and a preschool attended by children with normal development. Prior to the study, the Gilliam Autism Rating Scale-2- Turkish Version (GARS-2-TV), developed in 1995 by Gilliam and adapted for use in Turkey and standardized by Diken, Ardıç, and Diken (2012), was administered to all the subjects.

Prerequisite skills for teaching the target skills to the subjects are explained in the following. As a prerequisite, subjects were required to have the ability to (a) understand and perform verbal and visual instructions that involve at least two actions, (b) follow visual, verbal and auditory instructions, (c) direct attention to the images on the smartboard screen and watch the images for at least 2 minutes, (d) imitate verbal skills that involve at least two words, (e) direct attention to an event or situation for at least 5–6 minutes, (f) make eye contact, and (g) imitate facial expressions. To decide whether the subjects had these skills, first, meetings were held with their teachers and the subjects were observed in class. Then, the researcher tested the subjects for each skill.

Efe is 3 years and 9 months old. On the Stanford Binet Intelligence Test administered by an expert, Efe received an intelligence quotient score of 114. Moreover, he received an autistic disorder score of 77 on the GARS-2-TV administered. This score indicates that Efe is likely to have autistic disorder.

Uzay is 4 years and 3 months old. On the Stanford Binet Intelligence Test administered by an expert, Uzay received an intelligence quotient score of 130. Moreover, he received an autistic disorder score of 103 on the GARS-

2-TV administered. This score indicates that Uzay is highly likely to have autistic disorder.

Eren is 5 years and 3 months old. On the Stanford Binet Intelligence Test administered by an expert, Efe received an intelligence quotient score of 98. Moreover, he received an autistic disorder score of 86 on the GARS-2-TV administered. This score indicates that Efe is highly likely to have autistic disorder.

All subjects in the study were able to follow instructions that involve two or three actions, and speak in sentences that consist of a minimum of three or four words. The subjects have deficits in initiating communication, and are unable to use verbal language appropriate for the context. The subjects are comparable to their peers in terms of gross motor skills and fine motor skills.

Models. One female and two male adult models have participated in the study. Adult models took part in the study to demonstrate the social response behaviors (facial expressions, gestures and verbal expressions) that were necessary for creating the social context in the instructional videos to be watched by the subjects. Care was taken to ensure that models in the videos have as much diversity as possible in terms of age, gender and physical characteristics (weight, height, hair color and hair type). This is because the more diverse stimulus and response examples in the instruction process are, the more likely the subjects are to generalize the newly acquired skills without further instruction (Alberto & Troutman, 2013; Sulzer-Azaroff & Mayer, 1991).

Models in the study participated in a 2-hour information session about the research process and expectations from them in this process. This information session proceeded as follows: (a) explaining the aims of the study, (b) presenting discriminative stimuli that are appropriate for the context and that elicit social response behaviors, and explaining and demonstrating the proper social response behavior elicited by the discriminative stimulus, (c) explaining and demonstrating what the models need to pay attention to when shooting the videos, and (d) the researcher acting as a model and allowing the models to practice.

To establish the validity of the instructional videos, an Instructional Video Validity Form was prepared. Two experts who have worked on autism spectrum disorder and video modeling were shown the videos, and their opinions were obtained about the appropriateness of the videos. Experts were asked to evaluate the videos in terms of whether the videos contained social contexts related to affective states of happiness and anger, whether facial expressions were appropriate, and whether the images and audio were clear. Both experts were of the opinion that the videos were fit for the aims of the study.

Settings

Video modeling and probe sessions were conducted in one of the individual education rooms of the Developmental Support Unit of Anadolu University's Research Institute for Individuals with Disabilities. The rooms are equipped with observation mirrors, a rectangular table, two chairs, a closet for storing class materials, shelves and a smartboard.

Generalization probe sessions were held in group training rooms, a different setting from the rooms where instruction took place.

Materials

Materials used during the study were as follows: (a) a digital video camera, (b) a laptop computer for editing the videos, (c) video recordings with instructional content, (d) a smartboard, (e) objects and toys used to elicit the expected social response behavior from the subjects, and (f) data collection forms.

Studies show that preparing individual videos for each child is important in video modeling (Charlop et al., 2010; Murray & Noland, 2013). Therefore, individual videos were prepared using the favorite toys and activities of each child, and taking their individual characteristics into account. To identify the favorite activities and toys of the children, the researcher had meetings with the parents and teachers of the subjects, and observed them in class. The researcher also held preference evaluation sessions for the subjects to identify their preferences. Table 1 reports materials and activities identified for each subject based on information received from teachers and families and from preference evaluation sessions informed by observations.

Trainer

The trainer of the study was the first author of the manuscript. The trainer is currently working toward a PhD in special education. The trainer also has experience providing one-on-one and group training to children with autism.

Experimental Design

This study uses multiple probe design across participants, which is a single-subject design. In multiple probe design across participants, experimental control is achieved when the subject instructed starts giving more correct responses, no significant change is observed in the responses of the subjects not yet instructed, and this effect is consecutively repeated in the other subjects (Kennedy, 2005). To achieve experimental control, certain measures were taken to control factors that might affect internal validity. To control external factors prior to the study, the researcher informed the families and the teachers at the institution attended by the subjects about the study, and asked them not to provide training concerning the target behavior. In addition, the study was completed in a reasonable amount of time to control for maturity effect, and reliability data were collected in at least 20% of all sessions throughout the study to control measurement effects.

Dependent variable. The dependent variable of the study was defined as the performance of the appropriate social response behavior by the subject within 5 seconds of the emergence of a discriminative stimulus that elicits social response behaviors in the form of verbal expressions, gestures or facial expressions indicative of the affective states of happiness and anger. A literature review conducted to identify the dependent variable of the study showed that social response behavior was defined as facial expressions, gestures, verbal expressions and intonations that are used in the presence of a discriminative stimulus that elicits appropriate social response behavior (Charlop et al., 2010).

Six basic universal emotions (happiness, anger, sadness, fear, surprise, pain) are known to play a role in the comprehension and expression of emotions appropriate for the context (Ekman, 1999). This study, however, focuses on the emotions of happiness and anger. These emotions were selected because it is considered that preschool children with normal development, particularly in their interaction with peers in daily life, frequently use expressions of happiness and anger. The selection of the independent variable to be taught to the subjects was also informed by meetings held with the families and teachers of the children, the researcher's in-class observations, and preference evaluation conducted by the researcher.

Target responses and discriminative stimuli. Discriminative stimuli were identified on the basis of the objects, toys and activities most preferred by the subjects according to preference evaluation. Table 1 reports the discriminative stimuli selected for each subject. After identifying the discriminative stimuli, the researcher also prepared scenarios concerning social response behaviors that correspond to the affective states of happiness and anger elicited in the presence of each discriminative stimulus. Scenarios involving the discriminative stimuli and the target social response behaviors are presented in Table 2.

Independent variable. The independent variable in this study is video modeling presented via smartboard. Video modeling was provided on a one on one basis in two training sessions a day, 5 days a week.

Experimental Procedures

Experimental procedure of the study consisted of probe (baseline and daily probe sessions), instruction, maintenance and generalization sessions.

Probe sessions. There were two types of probe sessions in the study: baseline probe sessions and daily probe sessions.

Baseline probe sessions. Prior to initiating the teaching of the social response behavior to be studied, a minimum of three baseline probe sessions were held with each subject until stable data were obtained. With the first subject, stable data were obtained for three sessions in a row prior to initiating the instruction. With the other subjects who had not started receiving instruction, baseline data continued to be collected on an intermittent basis (once every 4 days). After the first subject

TABLE 1

Discriminative Stimuli Selected for Each Subject

	Discriminative Stimuli					
Subject	*Happiness*			*Anger*		
Efe	Knocking down pins with a bowling ball	Catching fish with a fishing rod	Attaching parts of the potato head to appropriate places	Taking the his iPad away when he was playing with it	Taking his car away when he was playing with it	Taking his story book away when he was looking at it
Uzay	Knocking down pins with a bowling ball	Playing darts	Catching fish with a fishing rod	Taking the his iPad away when he was playing with it	Taking his car away when he was playing with it	Taking his play-doh away when he was playing with it
Eren	Knocking down pins with a bowling ball	Playing darts	Catching fish with a fishing rod	Taking the his iPad away when he was playing with it	Taking his car away when he was playing with it	Taking his story book away when he was looking at it

achieved a 100% correct response rate for the behavior taught in three consecutive sessions, baseline data were collected from the second subject until stable data were obtained in three consecutive sessions. After the second subject achieved a 100% correct response rate in three consecutive sessions, the same procedure was repeated for the third subject.

In baseline sessions, a total of six trials were made for each subject, consisting of three discriminative stimuli for each of the affective states of happiness and anger, to give the subjects the opportunity to display the target behavior. The intertrials intervals were five seconds. In all probe sessions, trials were presented in an unpredictable order. In probe sessions, correct responses were reinforced using verbal reinforcements (e.g. Great!), following a continuous reinforcement schedule. Incorrect responses, on the other hand, were ignored. Focused and cooperative participation of the subject in the session was reinforced at the end of each session using verbal and social reinforcers (e.g. you worked great today, you were amazing. High five!).

When collecting data for the probes, scenarios (Table 2) developed for evaluating the behaviors to be taught were acted out. The trainer initiated the interaction with the subject by presenting one of the discriminative stimuli to the subject (e.g. giving an iPad for the subject to play with for a while, and then taking it back) to elicit the appropriate social response behavior. When the subject displayed the appropriate social response behavior within 5 seconds (e.g. frowning with an angry facial expression, shaking his finger at the researcher and saying "Don't take."), the subject was verbally reinforced, and incorrect responses were ignored.

Daily probe sessions. Daily probe sessions were held to measure the performance of the subjects concerning all the skills being taught during the instruction process, and data from these sessions are reported on the graph, marked as instruction phase data. Training sessions continued until the subjects achieved a correct response rate of 100% in three consecutive daily probe sessions. Daily probe sessions were held following every two training sessions, and once prior to training sessions held every day. In daily probe sessions, correct responses by the subjects were reinforced us-

TABLE 2

Training/Generalization Sessions in Which Discriminative Stimuli were Presented, and the Target Social Responses

Discriminative Stimuli for "Anger"		*Target Appropriate Social Responses*		
Training Scenarios	*Generalization Scenarios*	*Verbal Expression*	*Facial Expression*	*Gesture*
1. The trainer places an iPad on the table, and says "Let's sit on this chair and play with the iPad," addressing the subject. The subject sits, the trainer hands the iPad to the subject and then takes the iPad away after the subject plays games for a few minutes.	1. The trainer places a (chocolate) cake on the table, and says "Let's sit on this chair. Would you like some cake?", addressing the subject. The subject sits, the trainer hands the cake to the subject and then takes the cake away as the subject tries to unwrap the cake.	"Give it back" "Give it to me" "Let me have it"	Frowning, scowling, pursing lips.	Extending hands to the person who took the object away, with palms facing upwards or inwards.
2. The trainer places the toy car (a remote-controlled car or one with lights and sounds) on the table, and says "Let's sit on the chair and play with car. Let's drive the car," addressing the subject. After the subject plays with the car for a couple of minutes, the trainer takes the remote or the car away.	2. The trainer places an empty piece of paper and crayon on the table, and says "Let's sit on the chair and draw a picture," addressing the subject. The subject sits, the trainer hands the crayon to the subject and then takes the crayon away as the subject draws a picture.	"Don't take it" "Don't take, give it to me" "I said, give it to me"		Extending hands to the person who took the object away, with palms facing downwards, and pulling at their arms
3. The trainer places two story books on the table, says "Let's sit on the chair and read some books," addressing the subject, and asks "Which one would you like to read?", pointing to the books. The subject chooses a book, and the trainer gives the book to the subject. After the subject spends a couple of minutes with the book, the trainer takes the book away.	3. The trainer places two toys (sliced apple and cash register) on the table, says "Let's sit on the chair and play," addressing the subject, and asks "Which one would you like to play with?", pointing to the toys. The subject chooses a toy, and the trainer gives the toy to the subject. After the subject spends a couple of minutes with the toy, the trainer takes the toy away.	"No, don't take" "No, give it back"		Extending one hand to the person who took the object away, pointing with the index finger to the object.
4. The trainer places play-doh on the table, and says "Let's sit on this chair and play with the play-doh," addressing the subject. After the subject plays with the play-doh for couple of minutes, the trainer takes the play-doh away.				

TABLE 2—(Continued)

Discriminative Stimuli for "Happiness"		*Target Appropriate Social Responses*		
Training Scenarios	*Generalization Scenarios*	*Verbal Expression*	*Facial Expression*	*Gesture*
1. The trainer and the subject enter the room. The trainer says "Let's play bowling, but first we need to set up the pins," addressing the subject. The trainer and the subject together set up the pins, the trainer hands the bowling ball to the subject and says "We will stand by this line, and throw the ball to the pins. Let's go, you throw first." The subject takes the ball and knocks down the pins.	1. The trainer and the subject enter the room. The trainer says "Let's play football, but first we need to set up a goal post," addressing the subject. The trainer and the subject together set up the goal post, the trainer hands the ball to the subject, and says "We will stand by this line, and send the ball to the goal post. Let's go, you kick first." The subject takes the ball sends it to the goal post.	"Yay, I did it." "I made it." "I won." "Wow, I made it."	Raising the eyebrows, crinkling the eyes, baring the teeth	Making one hand into a fist, raising the fist, and moving the fist upwards and downwards
2. The trainer and the subject enter the room. The trainers says "Let's play darts," addressing the subject. The trainer hands the darts to the subject, places the dartboard on the table, and says "We will throw darts to the dartboard, and try to score 100. You go first, let's see if you can score a 100." The subject takes the darts, and throws them at the 100-point section.	2. The trainer and the subject enter the room. The trainer says "Let's play ring toss," addressing the subject. The trainer hands the rings to the subject, and says, "We will stand by this line and try to throw the rings onto the stake. You go first, let's see how many rings you can land on the stake," pointing to the line. The subject takes the rings and tosses them at the stake.	"Wow, I won." "Yes, I made it." "Yes, I won."		Making both hand into fists, raising the fists, and moving the fists upwards and downwards
3. The trainer and the subject enter the room. Once in the room, the trainer says "Let's sit on the chair, we will play fishing. Let's see if you can catch any fish.", addressing the subject. The subject sits at the table and fishes.	3. The trainer and the subject enter the room. The trainer places two bingo cards on the table, says "Let's sit on the chair and play bingo," addressing the subject, and asks "Which card would you like?", pointing to the cards. The subject sits at the table and chooses a card, and the trainer says "Let's see if you can place all the pictures in the right places." The subject places all the pictures in the bingo game in the right places.			Opening hands and arms so that palms face one another, and clapping
4. The trainer and the subject enter the room. Upon entering the room, the trainer says "We will play with Mr. Potato Head. Let's see if you can attach all the parts." The subject sits at the table and attaches all the parts correctly.				

ing verbal reinforcers (e.g. "Well done."), following a continuous reinforcement schedule. The procedure followed in baseline probe sessions was followed in daily probe sessions as well.

Training sessions. Training sessions were held twice a day, for 5 days a week. At least 45 min. had been interrupted between the two training sessions. Training sessions were held on a one-on-one basis. There were a total of six training videos, three for each of the affective states of happiness and anger. A total of six trials were made in each training session. Once the subjects met 50% of the criteria in training, trials in videos started to be presented in an unpredictable order in each training session. The intertrials intervals were 5 seconds.

In training sessions, the trainer and the subject sat side by side in front of the smartboard. Then, the trainer made a brief comment about the video in order to attract the attention of the subject to the video (e.g. "Now, we will watch a video with you. Look, there are two people. Let's see what they are going to do."). Then, the trainer played the video, and presented the target stimulus (e.g. "Uzay, please watch the video."). The trainer had the subject watch all six trials in the video without any intervals. If the subject got distracted when watching the video, the researcher directed the attention of the subject to the images on the smartboard by giving an instruction (e.g. "Uzay, please watch the video carefully."). When the subject watched the video carefully and made the correct responses, he was verbally reinforced.

Immediately after watching the video, the trainer created a setting in which the subject would have a chance to implement what the saw on the video. To attract the attention of the subject to the practice, the trainer presented discriminative stimuli (e.g. "Now we are going to play the game you have watched on the video, alright?"). The trainer initiated the interaction with the subject by presenting one of the discriminative stimuli to the subject (e.g. giving an iPad for the subject to play with for a while, and then taking it back) to elicit the appropriate social response behavior (facial expression, gesture and verbal expression). When the subject displayed the appropriate social response behavior within 5 seconds (e.g. frowning with an angry facial expression, shaking his hand at the researcher and saying "Don't take."), he was reinforced using verbal reinforcers (e.g. "Well done, you are great!"), following a continuous reinforcement schedule. In addition, because all of the discriminative stimuli used in training sessions were selected from among the favorite activities, games or toys of the subject, these discriminative stimuli also served as natural reinforcers for the subjects. Incorrect responses, on the other hand, were ignored. Focused and cooperative participation of the subject in the session was reinforced using verbal and social reinforcers at the end of the training session.

Maintenance and generalization sessions. Maintenance sessions were held 4 weeks, 6 weeks and 7 months after the criteria for target behavior were met. Pretest and posttest measures were used to evaluate whether the subjects generalized the skills that they have acquired. Generalization was evaluated with different people, settings, and contexts that were created using materials not included in the training sessions for teaching the target behavior. The procedure followed in probe sessions was also followed in maintenance and generalization sessions. Differently from probe sessions, reinforcement was faded, and focused participation by the subject was reinforced using verbal and social reinforcements at the end of the session only.

Social validity. A "Social Validity Questionnaire" was developed to examine the appropriateness of the methods used to teach the target behavior, and the significance of the results for the teachers and mothers of the subjects. Social validity data were collected using the subjective evaluation approach. The questionnaire for families contained a total of 10 questions. Of those, eight were closed ended and two were open ended questions. The questionnaire for teachers contained a total of nine questions, of which seven were closed-ended questions and two were open ended questions. These forms were distributed to families and teachers in closed envelopes, and collected without asking the participants to identify themselves. Data from the questionnaires were examined using descriptive analysis (Schwartz & Baer, 1991).

Reliability. Inter-observer reliability and treatment integrity data were collected in at

least 20% of the sessions held during the experimental stage the study. Reliability data were collected by the trainer and by an independent observer, a PhD student in special education. Inter-observer reliability data were analyzed using the following formula: [(Agreement/ Agreement + Disagreement) × 100] (Alberto & Troutman, 2013). Inter-observer agreement was 99.3%.

Treatment integrity data were analyzed using treatment integrity percentage, calculated using the following formula: "Observed trainer behavior / Planned trainer behavior × 100" (Billingsley, White, & Munson, 1980). Treatment integrity data for the training sessions were collected on the basis of the following behaviors: (a) preparing the video to be played on the smartboard, (b) preparing the materials and arranging the room, (c) presenting the attention prompt for watching the video, (d) presenting the target stimulus (e.g. "Please watch the video."), (e) having the subject watch the video, (f) reinforcing the watching of the video by the subject, (g) inviting the subject to the play area, (h) attracting attention by initiating interaction in an appropriate manner, (i) presenting the discriminative stimulus (toy or activity), (j) waiting for the response period (5 seconds), (k) reacting in an appropriate manner to the subject's response, (l) having an interval between trials, and (m) reinforcing the subject's participation. Treatment integrity was 98.6%.

Results

Effectiveness Results

An examination of the data on social response behaviors concerning the affective states of happiness and anger, target behaviors for Efe, Uzay and Eren to learn, showed that they did not display any correct responses in the initial stages. In the training stage, when video modeling was presented via smartboard, Efe, Uzay and Eren were observed to have a correct response rate of 100%. Data from maintenance sessions 4 weeks, 6 weeks and 7 months after the training showed that Efe and Uzay maintained the social response behaviors they learned with a 100% correct response rate, and Eren with an 83% correct response rate (see Figure 1).

Generalization Results

Generalization data were collected with different people, settings, and contexts that were created using materials not included in the training sessions for teaching the target behavior. Generalization data showed that Efe, Uzay and Eren were unable to display correct response behaviors in pretest generalization sessions. Data from the posttest session, on the other hand showed that Efe and Uzay were able to generalize their skills with a correct response rate of 100%, and Eren with a correct response rate of 83 % (see Figure 1).

Instructional Data

Data from the training sessions showed that Efe and Eren learned the social response behaviors concerning the affective states of happiness and anger after 18 sessions, and Uzay learned them after 14 sessions. A total of 108 trials were made in training sessions with Efe and Eren until the criteria were met, and 84 trials in the case of Uzay. Training sessions with Efe lasted 3 hours, 57 minutes and 3 seconds in total, 2 hours, 31 minutes and 7 seconds with Uzay, and 3 hours, 49 minutes and 55 seconds with Eren. Efe and Eren acquired the target social response behavior after nine daily probe sessions, and Uzay after seven daily probe sessions. A total of 54 trials were made in daily probe sessions with Efe and Eren, and 42 trials with Uzay. Training sessions with Efe lasted 1 hour, 15 minutes and 1 second in total, 40 minutes and 41 seconds with Uzay, and 1 hour, 9 minutes and 51 seconds with Eren. Before reaching the criteria for the target behavior in training sessions, Efe gave 53 (49.07%) incorrect responses, Uzay gave 40 (44.61%) incorrect responses, and Eren gave 58 (53.70%) incorrect responses. In daily probe sessions, on the other hand, Efe gave 15 (27.7%) incorrect responses, Uzay gave 18 (42.85%) incorrect responses, and Eren gave 22 (40.74%) incorrect responses (see Table 3).

Social Validity Results

Parents' opinions about the social validity of the study. Results of the study showed that all families were of the opinion that teaching

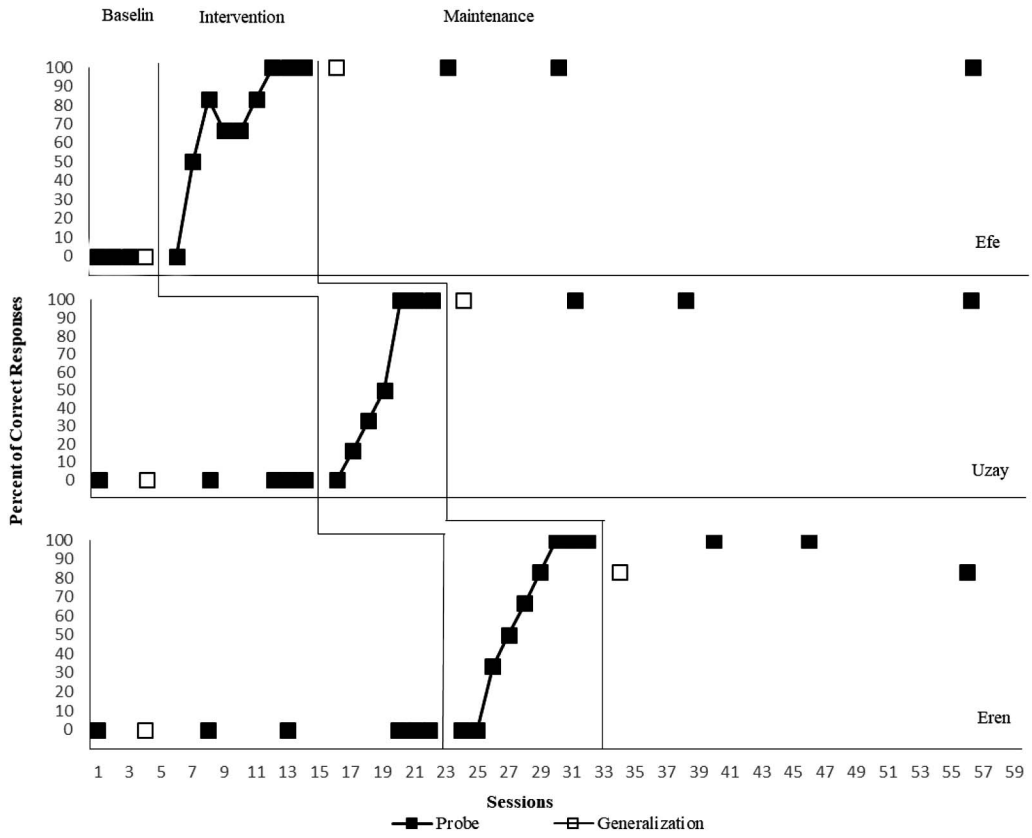


Figure 1. Percentage of correct responses in baseline, training, maintenance and generalization sessions.

social response behaviors in the form of verbal expressions, gestures and facial expressions was very important for their children, and video modeling presented via smartboard in the teaching of these behaviors was effective. When talking about the aspects of the study that they liked, parents said that their children repeat what they learn at school in non-school contexts as well. Families talked about the changes they observed in their children after the training as follows: *"He talks about his feelings in out activities, for example when we go to a park or mall." "He started saying 'I did it!' after visiting the bathroom." "He used to remain silent when his toy was taken away, now he started saying 'No, don't take!"*

Teachers' opinions about the social validity of the study. All of the teachers said that teaching social response behaviors to their students and the use of video modeling presented via

TABLE 3

Instructional Data on Social Response Behaviors

Subjects	*No. of Training Sessions/Trials*	*No. and % of Training Errors*	*Training Time h:m:s*	*No. of Daily Probe Sessions/Trials*	*No. and % of Daily Probe Errors*	*Daily Probe h:m:s*
Efe	18/108	53–49, 07%	03:57:03	9/54	15–27, 7%	01:15:01
Uzay	14/84	40–47, 61%	02:31:07	7/42	18–42, 85%	00:40:41
Eren	18/108	58–53, 70%	03:49:55	9/54	22–40, 74%	01:09:51

smartboard to this end were important for the education of their students. When talking about the aspects of the study that they liked, teachers said that they observed the children repeat what they learned in the classroom.

Discussion

This study examined the effectiveness of video modeling presented via smartboard for teaching children with autism about social response behaviors concerning the affective states of happiness and anger. In addition, social validity data in the form of subjective evaluations were collected from the parents and teachers of the subjects who participated in the study. In the following sections, various aspects of the study are discussed in light of study results, and recommendations are made for future studies.

In the relevant literature, a limited number of studies were found on the teaching of social response behaviors concerning the affective states of happiness and anger (Akmanoğlu, 2015; Axe & Evans, 2012; Charlop-Christy & Daneshvar, 2003; Gena et al., 2005), and a single study that aimed to teach all social response behaviors in the form of gesture, intonation, verbal and facial expression (Charlop et al., 2010). Results of the present study are consistent with the results of other studies on the teaching of social response behaviors. However, the previous study aiming to teach all social response behaviors (Charlop et al., 2010), social response behaviors were planned and taught without associating these responses with affective states or contexts related to these affective states. In the present study, on the other hand, differently from the previous study, different affective states and different people associated with the social responses, identified as important variables in the emergence of social responses, were examined. One of the strengths of the present study is that it teaches all of the behaviors related to the main deficits of children with autism in a holistic manner and in relation to different contexts. In addition, the results of this study contributes to the limited literature on the teaching of this skill (Charlop et al., 2010).

Seven months after the instruction of the subjects who participated in the study was completed, data were collected to see whether the social response behaviors concerning the affective states of happiness and anger were maintained. Results of the study showed that, 7 months after the instruction, all of the subjects maintained the behaviors they have learned. Results of the present study concerning maintenance are consistent with those of previous studies on the teaching of various social response behaviors to children with autism (Gena et al., 1996; Gena et al., 2005). However, the one study on the teaching of social response behavior in the form of both verbal and non-verbal behaviors does not report maintenance data (Charlop et al., 2010). Therefore, results of the present study on maintenance makes a contribution to this limited literature.

Results of the study show that social response behaviors concerning the affective states of happiness and anger, taught using video modeling, is generalized to different people, settings and contexts that are created using different materials. Results of the study are thus consistent with those of other studies in the literature (Charlop et al., 2010). However, an examination of the data on generalization in other studies shows that these studies did not evaluate the target behaviors by creating different social contexts. The present study, on the other hand, has evaluated generalization with different people, settings and social contexts created using materials not included in training, and by taking into account probable scenarios that the subjects may encounter in daily life. Given the problems experienced by children with autism concerning generalization (Neisworth & Wolfe, 2005), generalization sessions constitute another strength of the present study. In addition, results of the present study concerning generalization expand the existing results reported in the literature.

In terms of social validity, results of the present study showed that mothers and teachers of the children who participated in the study had positive opinions of the study. Charlop et al. (2010) collected social validity data from undergraduate students. In the present study, on the other hand, social validity data are collected from parents and teachers. In other words, the two studies collected social validity data from different sources. Neverthe-

less, both studies report similar social validity results. Collecting social validity data exclusively from the subjects that are the direct participants in the study is an important limitation, and collecting social validity data from parents and teachers of the subjects removes this limitation (Schwartz & Baer, 1991). Moreover, very few studies on the exclusive use of video modeling to teach social skills report social validity data (Acar & Diken, 2012). Thus, another contribution of the present study to the literature is that it collects social validity data from both parents and teachers.

Results of the present study on effectiveness, generalization, maintenance and social validity are positive, but there are a number of additional points that need to be considered. First, subjects who participated in the study were observed to display gesture and verbal expression behaviors in a more natural manner, but their facial expressions were more mechanic and it took longer to acquire this skill compared to the other skills. It is thought that subjects with higher degrees of autism, indicated by their GARS-2-TV scores, had more unnatural facial expressions, in the form of simple imitation and displaying a more mechanic expression. Diagnosis criteria in the literature show that children with higher degrees of autism need more support, particularly in the fields of social communication and interaction (APA, 2013; Kırcaali-İftar, 2013). Results of the study are thus consistent with other results in the literature. Second, results from the training sessions showed that Uzay reached the target criteria in a shorter amount of time compared to the other subjects. Even though Uzay's GARS-2-TV score was higher than those of the other subjects, he also received a higher score on the Stanford Binet Intelligence, which might explain why he reached the target criteria faster. Deficiencies in the social affective development of children with autism are insufficient to explain affective deficits in autism; deficiencies in terms of cognitive development also hinder the social and affective development of children with autism (Baron-Cohen, 1988, 1989). Third, given that preparing individual videos is important in video modeling (Charlop et al., 2010; Murray & Noland, 2013), all of the discriminative stimuli used in training sessions were selected from among the favorite activities, games or toys of each subject. These discriminative stimuli, selected from among the favorite activities, games and toys of each subject, also served as natural reinforcers, which is among the strengths of the present study.

Limitations and Future Research

Results of the study in terms of reliability, effectiveness and social validity were positive, but the study had some limitations as well. First, of the social response behaviors, facial expression, gesture and verbal expression were studied, but intonation was not included in the study. Second, social response behaviors concerning only two affective states, those of happiness and anger, were studied. Third, an effort was made to create diverse and natural social contexts to elicit social response behaviors, but these contexts were not naturally occurring situations in daily life. A final limitation that training was not provided concerning cases where the trainer refuses to give back the object demanded by the child, even though he displays the social response behavior concerning the affective state of anger when his toy is taken away.

Future studies are recommended to hold training sessions to teach social response behaviors to autism with children in more natural settings. This study examined social response behaviors in the form of facial expressions, gestures and verbal expressions. Thus, future studies are recommended to include intonation among the social response behaviors taught. In addition, future studies can aim to teach social response behaviors concerning other emotions besides happiness and anger taught in the present study. Future studies could also provide training concerning cases where the trainer refuses to give back the object demanded by the child, even though he displays the social response behavior concerning the affective state of anger when his toy is taken away.

In conclusion, video modeling presented via smartboard was found to be an effective method for teaching children with autism about social responses in the form of gestures, verbal and facial expressions indicative of the affective state of happiness and anger. Researchers and practitioners working with individuals with autism are recommended to use

video modeling to teach social response behaviors and different affective states.

References

Acar, Ç., & Diken, İ. H. (2012). Reviewing instructional studies conducted using video modeling to children with autism. *Educational Sciences: Theory & Practice, 12,* 2719–2738.

Akmanoglu, N. (2015). Effectiveness of teaching naming facial expression to children with autism via video modeling. *Educational Sciences: Theory & Practice, 15,* 519–537.

Alberto, A. P., & Troutman, C. A. (2013). *Applied behavior analysis for teachers* (9th ed.). Upper Saddle River, NJ: Pearson Education.

American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders:* DSM-5 (5th ed.). Arlington, VA: Author.

Argott, B. (2012). *The effects of teaching using the smartboard versus discrete trial teaching on acquisition and student engagement for children with autism.* Ann Arbor, MI: Dissertation Publishing.

Axe, J. B., & Evans, C. J. (2012). Using video modeling to teach children with PDD-NOS to respond to facial expressions. *Research in Autism Spectrum Disorders, 6,* 1176–1185.

Balconi, M., Amenta, S., & Ferrari, C. (2012). Emotional decoding in facial expression, scripts and videos: A comparison between normal, autistic and Asperger children. *Research in Autism Spectrum Disorders, 6,* 193–203.

Balconi, M., & Carrera, A. (2007). Emotional representation in facial expression and script: A comparison between normal and autistic children. *Research in Developmental Disabilities, 28,* 409–422.

Banda, D. R., Copple, K. S., Koul, R. K., Sancibrian, S. L., & Bogschutz, R. J. (2010). Video modelling interventions to teach spontaneous requesting using AAC devices to individuals with autism: a preliminary investigation. *Disability and Rehabilitation, 32,* 1364–1372.

Baron-Cohen, S. (1988). Social and pragmatic deficits in autism: Cognitive or affective? *Journal of Autism and Developmental Disorders, 18,* 379–402.

Baron-Cohen, S. (1989). The autistic child's theory of mind: A case of specific developmental delay. *Journal of Child Psychology and Psychiatry, 30,* 285–297.

Billingsley, F. F., White, O. R., & Munson, R. (1980). Procedural reliability: A rationale and an example. *Behavioral Assessment, 2,* 2.

Buffington, D. M., Krantz, P. J., McClannahan, L. E., & Poulson, C. L. (1998). Procedures for teaching appropriate gestural communication skills to children with autism. *Journal of Autism and Developmental Disorders, 28,* 535–545.

Charlop, M. H., Dennis, B., Carpenter, M. H., & Greenberg, A. L. (2010). Teaching socially expressive behaviors to children with autism through video modeling. *Education and Treatment of Children, 33,* 371–393.

Charlop-Christy, M. H., & Daneshvar, S. (2003). Using video modeling to teach perspective taking to children with autism. *Journal of Positive Behavior Interventions, 5,* 12–21.

Charlop-Christy, M. H., Le, L., & Freeman, K. A. (2000). A comparison of video modeling with in vivo modeling for teaching children with autism. *Journal of Autism and Developmental Disorders, 30,* 537–552.

Delano, M. E. (2007). Video modeling interventions for individuals with autism. *Remedial and Special Education, 28,* 33–42.

Diken, İ. H., Ardıç, A., & Diken, Ö. (2012). *Gilliam Autism Rating Scale-2 Turkish Version.* Ankara: Maya Akademi Yayınları.

Ekman, P. (1999). Basic emotions. In T. Dalgleish & M. Power (Eds.), *Handbook of cognition and emotion* (pp. 45–60). Chichester: John Wiley & Sons.

Gena, A., Krantz, P. J., McClannahan, L. E., & Poulson, C. L. (1996). Training and generalization of affective behavior displayed by youth with autism. *Journal of Applied Behavior Analysis, 29,* 291–304.

Gena, A., Couloura, S., & Kymissis, E. (2005). Modifying the affective behavior of preschoolers with autism using in-vivo or video modeling and reinforcement contingencies. *Journal of Autism and Developmental Disorders, 35,* 545–556.

Ingersoll, B., Lewis, E., & Kroman, E. (2006). Teaching the imitation and spontaneous use of descriptive gestures in young children with autism using a naturalistic behavioral intervention. *Journal of Autism and Developmental Disorders, 37,* 1446–1456.

Kennedy, C. H. (2005). *Single-case designs for educational research.* Boston, MA: Prentice Hall.

Kırcaali-İftar, G. (2013). Otizm spektrum bozukluğuna genel bakış. In E. Tekin-İftar (Ed.), *Otizm spektrum bozukluğu olan çocuklar ve eğitimleri,* (Children with autism spectrum disorders and their education) (pp. 17–43). Ankara: Vize Basın Yayınları.

Leaf, J. B., Dotson, W. H., Oppeneheim, M. L., Sheldon, J. B., & Sherman, J. A. (2010). The effectiveness of a group teaching interaction procedure for teaching social skills to young children with a pervasive developmental disorder. *Research in Autism Spectrum Disorders, 4,* 186–198.

Murray, S., & Noland, B. (2013). *Video modeling for young children with autism spectrum disorders.* Philadelphia: Jessica Kingsley Publishers.

National Autism Center. (2015). *The national autism center's national standards report phase 2.* Massachusetts: National autism center. Retrieved from http://www.nationalautismcenter.org/reports

National Professional Developmental Center on Autism Spectrum Disorder. (2014). *Evidence-based practice brief components.* Retrieved from http:// auismpdc.fpg.unc. edu// sites/autismpdc.fpg.unc. edu/files/imce/documents/2014EBP-Report.pdf

Neisworth, J. T., & Wolfe, P. S. (2005). *The autism encyclopedia.* Baltimore, MD: Paul H Brookes Pub Co.

Nikopoulos, C., & Keenan, M. (2006). *Video modeling and behaviour analysis.* London: Jessica Kingsley Publishers.

Quill, K. A. (1997). Instructional consideration for young children with autism: The rationale for visually cued instruction. *Journal of Autism and Developmental Disorders, 27,* 697–714.

Plimley, L., & Bowen, M. (2007). *Social skills and autistic spectrum disorders.* London: Paul Chapman Publishing. Retrieved from https://books.google. com.tr/books.

Saymaz, E. B. (2008). Otizmde duygu algılama ve ifade etme: Bir olgu sunumu. *Çocuk ve Gençlik Ruh Sağlığı Dergisi, 15,* 32–36.

Schneider, N., & Goldstein, H. (2010). Using social stories and visual schedules to ımprove socially appropriate behaviors in children with autism. *Journal of Positive Behavior Interventions, 12,* 149–160.

Schwartz, I. S., & Baer, D. M. (1991). Social validity assessments: Is current practice state of the art? *Journal of Applied Behavior Analysis, 24,* 189–204.

Sulzer-Azaroff, B., & Mayer, G. R. (1991). *Behavior analysis for lasting change.* New York, NY: Holt, Rinehart & Winston.

Tardif, C., Laine, F., Rodriguez, M., & Gepner, B. (2007). Slowing down presentation of facial movements and vocal sounds enhances facial expression recognition and induces facial–vocal imitation in children with autism. *Journal of Autism and Developmental Disorders, 37,* 1469–1484.

Webber, J., & Scheuermann, B. (2008). *Educating students with autism: A quick start manual.* Austin, TX: Pro-ed.

Weiss, M. J. (2013). Behavior analytic ınterventions for developing social skills. In P. F. Gerhardt & D. Crimmins (Eds.), *Social skills and adaptive behavior in learners with autism spectrum disorders* (pp. 373–51). Baltimore, MD: Brookes.

Received: 2 August 2017
Initial Acceptance: 4 October 2017
Final Acceptance: 5 November 2017